UNIVERSITY of SAN CARLOS
SCIENTIA • VIRTUS • DEVOTIO

Department of Computer Engineering
School of Engineering

Internship Report Form

Student-Intern's Name	:	Skyler Jason P. Banzon
Name of HTE	:	Cubeworks Technology Consulting and Solutions, Inc.
HTE Supervisor's Name	:	Elmer N. Canque (CIO)
Report Period **From**	:	April 16, 2026 **To** : May 15, 2026
Number of Hours Worked This Quarter	:	176
Cumulative No. of Hours to Date	:	416

Quarter Summary

No.	Tasks Performed	% Time Spent
1	Sugar Volume Project: Developed depth-stream processing for Kinect v1 to extract 11-bit raw data at 640 x 480 resolution.	15%
2	Sugar Volume Project: Implemented bulk mass calculation using pixel-area integration and density constants (850 kg/m3).	15%
3	Wire Harness Project: Created a BranchTracker class using greedy nearest-neighbor matching to maintain stable branch identity.	12%
4	Wire Harness Project: Built a modular 1080p Tkinter interface with real-time tuning sliders and a detachable mask window.	13%
5	Wire Harness Project: Developed topology algorithms to differentiate between the main trunk and Y-shaped junctions.	12%
6	Wire Harness Project: Automated scaling and coordinate transformation via 4-point ArUco homography at 4.0 px/mm.	10%
7	General CV Optimization: Integrated temporal filtering and deque buffers to stabilize measurement jitter across both projects.	13%
8	General CV Optimization: Optimized the vision pipeline using CLAHE and bilateral filtering for robust detection in warehouse lighting.	10%
	TOTAL	100%

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Additional Activities (other growth opportunities e.g., conferences, training, seminars etc.)

- None

List any holidays, personal days, and/or days missed this quarter:

- Observed national holidays during the reporting period: Labor Day, we proceeded with work online during the special non-working holidays around the time of the ASEAN Summit in Mactan

Quarter Analysis, Reflection, and Plan of Action

What experiences were particularly rewarding during this reporting period?

Implementing real-time volumetric mass estimation using Kinect depth data and developing a stable branch tracking system provided a strong sense of technical achievement. Successfully transitioning the wire harness project into a modular 1080p Tkinter interface also streamlined the debugging process and improved the overall professional quality of the system.

What experiences were particularly difficult during this reporting period?

Filtering depth data noise and floor interference for the sugar mountain project required precise percentile-based calibration. Additionally, developing complex topology logic to accurately differentiate between main trunks and Y-shaped junctions in the wire harness project proved technically challenging

(If Applicable) Other comments, items, or issues that you want your supervisor or adviser to be made aware of? What is your proposed plan of action to resolve this issue?

No major issues to report

Skyler Jason P. Banzon
Student-Intern

Elmer N. Canque
HTE Supervisor

Date Submitted : May 15, 2026

Date Reviewed : May 15, 2026

Supervisor's Comments:

None